



Fig. 1 — Central HVAC System Overview

HVAC SYSTEMS DESIGN MANUAL

Load Calculations
Duct Design
Equipment Selection
Controls & Energy Efficiency

12+
Chapters

Key Systems
HVAC Topics

Formulas
Included

Industry
Standard



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HVAC Fundamentals

Thermodynamics, Heat Transfer & System Types

What is HVAC?

HVAC (Heating, Ventilation, and Air Conditioning) encompasses the technology and engineering disciplines used to achieve thermal comfort and acceptable indoor air quality. Modern HVAC systems serve residential, commercial, and industrial applications ranging from simple split units to complex central plant systems.

Core Thermodynamic Principles

- Heat Transfer Modes: Conduction, Convection, Radiation
- First Law: Energy conservation — $Q = mc\Delta T$ (sensible heat)
- Second Law: Heat flows from high to low temperature (driving principle of refrigeration)
- Latent Heat: Energy absorbed/released during phase change (evaporation, condensation)
- Specific Heat: Air = 1.006 kJ/kg·K; Water = 4.186 kJ/kg·K

KEY FORMULA — Sensible Cooling Load:
 $Q_s = 1.23 \times V \times \Delta T$ (Q in W, V in L/s, ΔT in °C)
KEY FORMULA — Latent Cooling Load:
 $Q_l = 3010 \times V \times \Delta \omega$ (ω = humidity ratio g/kg)

HVAC System Classifications

System Type	Configuration	Best Application	Typical COP
Split/Multi-split	DX Refrigerant	Small commercial / residential	2.5–4.5
VRF/VRV	Variable refrigerant flow	Medium commercial, offices	3.5–5.0
Fan Coil Units (FCU)	Chilled/hot water	Hotels, large commercial	—
Air Handling Units	Central ducted	Factories, hospitals	—
Chilled Water Plant	Central plant	Large commercial, campus	5.0–7.0
DX Rooftop (RTU)	Packaged self-contained	Retail, light commercial	2.8–4.2

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Psychrometrics

Air Properties, Moisture & Comfort Conditions

Psychrometric Properties of Moist Air

- Dry-Bulb Temperature (DBT): Sensible temperature measured by thermometer
- Wet-Bulb Temperature (WBT): Temperature after adiabatic saturation — affects cooling tower design
- Dew Point: Temperature at which condensation begins — critical for chilled water coil design
- Relative Humidity (%RH): Ratio of actual to saturation vapour pressure
- Specific Humidity (ω): Mass of water vapour per kg of dry air (g/kg)
- Enthalpy (h): Total heat content of moist air (kJ/kg) — used in coil calculations

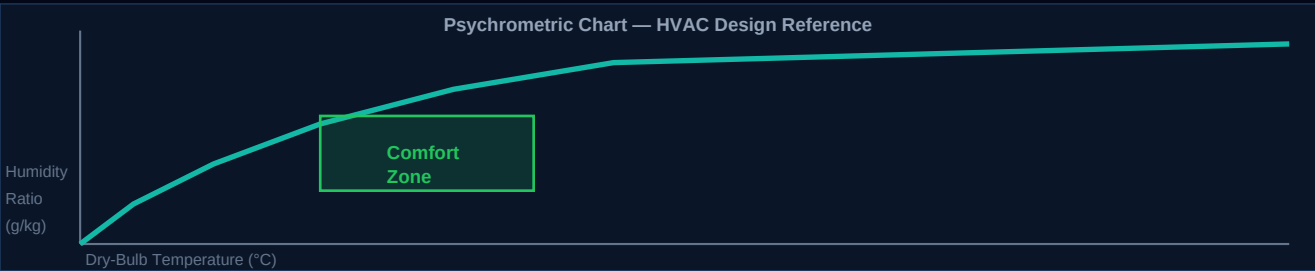


Fig. 2.1 — Simplified Psychrometric Chart

ASHRAE Comfort Conditions

- Summer: DBT 23–26°C | RH 30–60% | Air velocity < 0.25 m/s
- Winter: DBT 20–23°C | RH 30–50% | Air velocity < 0.15 m/s
- Operative Temperature: Average of air temp & mean radiant temp
- PMV (Predicted Mean Vote): Target range -0.5 to +0.5

Property	Symbol	Unit	Typical Range
Dry-Bulb Temp	DBT	°C	20–45 (outdoor)
Wet-Bulb Temp	WBT	°C	5–35
Relative Humidity	RH	%	30–70 (indoor)
Specific Humidity	ω	g/kg	8–20
Enthalpy	h	kJ/kg	40–100
Specific Volume	v	m³/kg	0.82–0.96

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Load Calculations

Heating & Cooling Load Estimation Methods

Cooling Load Sources

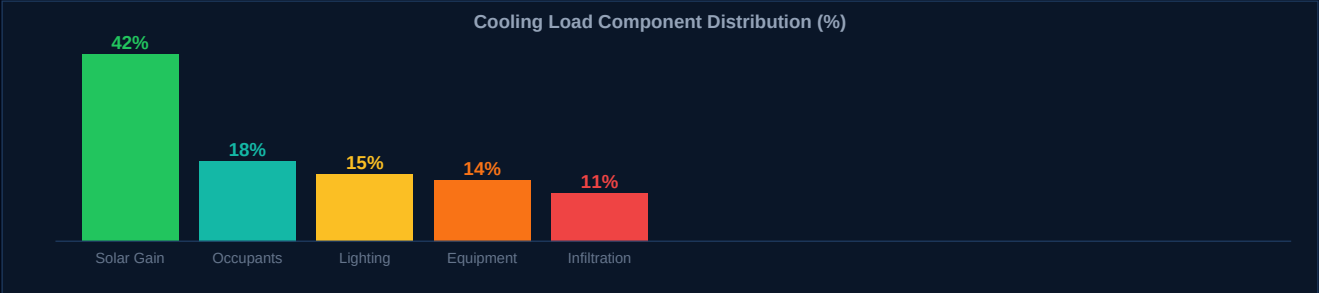


Fig. 3.1 — Cooling Load Component Breakdown

Heat Gain Components

- Solar Heat Gain (SHGC): $Q_{solar} = A \times SHGC \times SC \times CLF$
- Transmission: $Q_{trans} = U \times A \times (T_{out} - T_{in})$ [W]
- Infiltration Sensible: $Q_{inf} = 0.33 \times n \times V_{room} \times \Delta T$
- Occupants: Sensible 75W + Latent 55W per person (office activity)
- Lighting: $CLF \times Watts \times 3.412$ [BTU/hr] or $\times 1.0$ [W]
- Equipment: 70–100% of nameplate rating (diversity factor applied)

RULE OF THUMB — Preliminary Sizing:
 Offices: 80–120 W/m² | Retail: 120–180 W/m²
 Hotels: 60–90 W/m² | Hospitals: 100–150 W/m²
 Data Centres: 1000–3000 W/m² (server-driven, not envelope)
 Safety factor: add 10–15% to calculated load

Heating Load Calculation

- Fabric Losses: $Q_{fab} = \Sigma U \cdot A \times \Delta T_{design}$
- Ventilation Losses: $Q_{vent} = 0.33 \times ACH \times V \times \Delta T$
- Design ΔT = Indoor setpoint (21°C) – Outdoor design temp
- ASHRAE 99.6% design dry-bulb: site-specific (check ASHRAE HOF)
- No solar/internal gains credited for heating load (conservative)

Space Type	Cooling W/m²	Heating W/m²	Ventilation L/s·p
Open Plan Office	90–120	40–60	10
Conference Room	100–150	50–70	10
Server Room	600–3000	N/A	per heat load
Kitchen (Commercial)	200–400	80–120	35–50
Retail	120–180	60–90	10
Hospital Ward	100–140	60–80	20–30

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Air Distribution

Duct Design, Sizing & Air Terminal Devices

Duct Design Methods

- Equal Friction Method: Constant pressure loss per metre (0.8–1.5 Pa/m). Most common.
- Velocity Reduction: Step-down velocity from main to branches. Simple but less accurate.
- Static Regain: Pressure recovery in transitions — used for long low-velocity systems.
- T-Method (Optimization): Minimises lifecycle cost. Best for large complex systems.

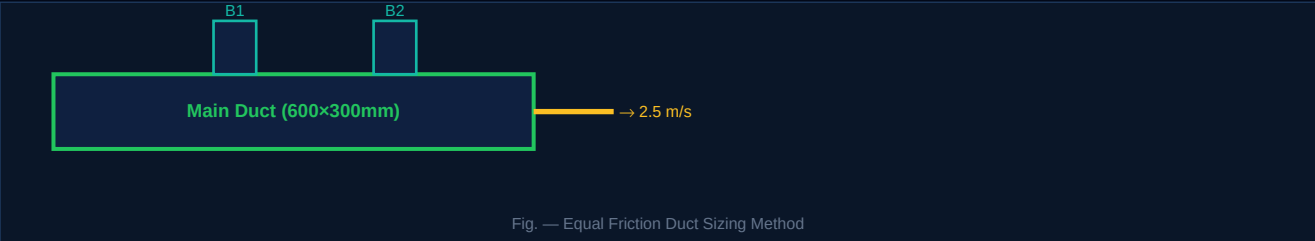


Fig. — Equal Friction Duct Sizing Method
 Fig. 4.1 — Equal Friction Duct Sizing & Branch Configuration

Recommended Air Velocities

Duct Section	Velocity (m/s)	Pressure Loss (Pa/m)	Notes
Main supply duct	5.0–8.0	0.8–1.5	Low noise, offices
Branch ducts	3.0–5.0	0.8–1.2	Keep < 5 for quiet
Final runouts	2.0–3.0	0.5–0.8	Last metre to terminal
Supply grille face	2.0–2.5	—	Controls throw & noise
Return grilles	1.5–2.0	—	Less critical than supply
High velocity (IU)	10–15	—	Industrial, not comfort

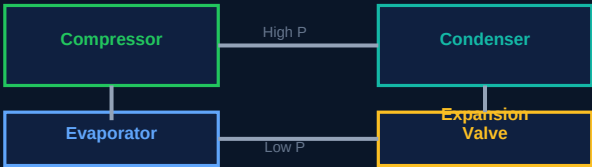
Air Terminal Devices

- Ceiling Diffusers: 360° spread, ideal for open-plan; 1 per 15–25 m² typical
- Linear Slot Diffusers: Wall/floor perimeter installations; good for floor-to-ceiling glass
- Swirl Diffusers: Used for high ceilings (>4m) and displacement ventilation
- Grilles & Registers: Simple supply/return — low cost, directional control via louvers
- Induction Units: High-velocity primary air induces room air; used in perimeter zones

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Mechanical Cooling

Chillers, DX Systems & Refrigeration



Vapour Compression Refrigeration Cycle
 Fig. 5.1 — Vapour Compression Refrigeration Cycle

Chiller Types & Selection

Chiller Type	Capacity Range	COP	Refrigerant	Best For
Air-Cooled Scroll	20–400 kW	2.5–3.5	R410A / R32	Small-medium buildings
Air-Cooled Screw	200–1500 kW	3.0–4.0	R134a / R513A	Medium commercial
Water-Cooled Centrifugal	500–5000+ kW	5.5–7.0	R134a / R1234ze	Large central plant
Water-Cooled Screw	200–2000 kW	4.5–6.0	R134a	Versatile medium–large
Absorption (LiBr-H ₂ O)	300–5000 kW	0.7–1.2 (COP _{t/h})	Water (refrigerant)	Waste heat / trigeneration

Refrigerant Transition — Phase-Out Schedule

- R22 (HCFC): Phased out globally — service only from recovered stock
- R410A (HFC): High GWP=2088 — EU F-Gas Phase-down by 2030
- R32 (HFC): GWP=675 — transitional, widely used in split systems
- R134a: GWP=1430 — phasing out in new equipment by 2025 (EU)
- R1234ze/R1234yf (HFO): GWP<1 — emerging standard for new chillers
- CO₂ (R744): Natural refrigerant, GWP=1 — transcritical systems growing

COOLING TOWER SELECTION:

Approach Temp = Leaving Water Temp – Wet-Bulb Temp (target: 3–4°C)

Range = Entering – Leaving CW Temp (typically 5–6°C)

Size on L/min per kW: approx. 3.6 L/min per kW of condenser heat rejected

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Heating Systems

Boilers, Heat Pumps & Radiant Heating

Boiler Types & Efficiency

Boiler Type	Efficiency (%)	Fuel	Output Temp	Application
Non-condensing	80–85	Gas/Oil	70–90°C	Older systems, high temp
Condensing	90–98	Gas	55–80°C	Modern commercial & residential
Electric Immersion	99	Electricity	Variable	Small / backup
Biomass Pellet	78–85	Wood Pellets	70–85°C	Sustainable / rural
Combined Heat & Power	CHP 80–90%	Gas/Bio	80–95°C	Large sites, simultaneous

Heat Pump Technology

- Air-Source Heat Pump (ASHP): COP 2.5–4.5 at 7°C ambient, 35°C flow temp
- Ground-Source Heat Pump (GSHP): COP 3.5–5.5 — stable ground temp (8–12°C)
- Water-Source HP: COP 4.0–6.0 — rivers, lakes, wastewater loops as heat sink
- Coefficient of Performance (COP) = Heat Output / Electrical Input
- SCOP (Seasonal): Average COP across heating season — used for EU energy ratings
- Low Temperature Systems (35–45°C) maximise HP efficiency; ideal with underfloor heating

HEAT PUMP SIZING RULE:
Capacity = Peak Heat Loss (W) ÷ [1 - (1/COP)] ← electrical input portion
Minimum outdoor design temp: ASHP rated to -15°C (cold climate) / -7°C (standard)
Buffer tank recommended: ≥50L per kW for cycling protection

Radiant Heating & Underfloor Systems

- UFH Pipe Spacing: 100–200mm centres; flow temp 35–45°C (wet) or 55–65°C (screed)
- Surface Temperature Limit: Max 29°C occupied areas; 35°C bathrooms
- Heat Output: 80–120 W/m² typical; floor construction U-value affects performance
- Radiant Panels: Ceiling/wall; 45–60°C water; fast response, no ductwork
- Infrared (IR) Heaters: Industrial/warehouse; radiant efficiency 85–95%

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Ventilation & IAQ

ASHRAE 62.1, Fresh Air & Pollutant Control

Ventilation Rate Requirements

Space Type	Rp (L/s·p)	Ra (L/s·m²)	Typical OA (L/s·p)
Office — Open Plan	5	0.3	10
Conference Room	5	0.3	10
Retail Store	7.5	0.3	15
Restaurant Dining	7.5	0.6	18
Hospital Patient Room	—	—	25–30
Classroom	5	0.6	12
Gymnasium	5	0.3	10

Rp = People outdoor air rate (breathing zone) | Ra = Area rate (building-related pollutants)

Indoor Air Quality Pollutants

- CO: < 1000 ppm (ASHRAE) — indicator of ventilation adequacy
- CO: < 9 ppm (8hr TWA) — combustion gases, parking garage exhaust
- PM2.5: < 12 µg/m³ (EPA annual average) — requires MERV-13+ filtration
- VOCs (Total): < 300 µg/m³ — off-gassing from materials, cleaning products
- Formaldehyde: < 0.1 ppm — composite wood, insulation; address at source
- Legionella: Control via cooling tower management, temperature maintenance

DEMAND-CONTROLLED VENTILATION (DCV):

Modulate OA based on CO sensors (setpoint: 700-800 ppm above ambient)

Energy Savings: 20–40% OA reduction at partial occupancy

Required by: ASHRAE 90.1 for occupancy > 25 people / 40 m²

Heat Recovery Ventilation (HRV/ERV)

- Plate Heat Exchanger: 60–80% sensible efficiency; no cross-contamination
- Rotary Heat Wheel: 75–85% total efficiency; slight cross-leakage
- Run-Around Coil: Cross-contamination proof; 45–65% efficiency
- ERV (Energy Recovery): Transfers both heat AND moisture — ideal for humid climates
- Minimum HRV required by ASHRAE 90.1 when OA > 70% of supply

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Humidity Control

Humidification, Dehumidification & Mould Prevention

Humidification Systems

Type	Method	Energy	Hygiene Risk	Capacity Range
Steam (Electric)	Electrode boiler	High	Low	5–500 kg/hr
Steam (Gas)	Gas-fired boiler	Medium	Low	50–2000 kg/hr
Evaporative Pad	Adiabatic (water)	Low	Medium	5–200 kg/hr
Ultrasonic	Piezoelectric vibration	Low	Medium-High	0.5–50 kg/hr
High-Pressure Fog	Atomised nozzles	Low	Medium	10–500 kg/hr

Dehumidification Strategies

- Cooling Coil: Reduce air below dew point — most common method in AC systems
- Desiccant Wheel: Chemical absorption; regenerated with heat — hospitals, food processing
- Heat Pipe Assisted: Pre-cool before coil, reheat after — improves dehumidification efficiency
- Chilled Beam: Cannot handle latent loads — always use with separate OA ventilation

MOULD PREVENTION:

- Maintain RH < 70% (surface temp > dew point of room air)
- Cold bridge factor: fRsi > 0.75 (UK Building Regs) to prevent surface condensation
- Thermal bridges at junctions — calculate using software (THERM, PSI values)
- Vapour barriers on warm side of insulation in cold climates

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HVAC Controls

BMS, DDC, VFDs & Smart Building Integration

Control Strategies

- On/Off Control: Simple thermostat — acceptable for residential; poor energy performance
- Proportional (P): Output proportional to error — faster response, steady-state offset
- PID Control: Proportional + Integral + Derivative — eliminates offset, standard for AHU
- Setpoint Reset: Adjust supply air/water temp based on zone demand — saves energy
- Optimal Start: Learn thermal mass & pre-condition space before occupancy

Variable Speed Drives (VSD/VFD)

AFFINITY LAWS (Fan / Pump):

Flow ∝ Speed | Pressure ∝ Speed² | Power ∝ Speed³

Reducing speed to 80% → Power reduces to 51% (0.8³)

Energy Savings: 30–50% vs. constant speed with damper control

Apply to: Supply/Return fans, Chilled water pumps, Condenser water pumps

Building Automation System (BAS/BMS)

Protocol	Application	Communication	Speed
BACnet	HVAC / Fire / Lighting	IP, MS/TP, LON	High
Modbus RTU/TCP	Chillers, VFDs, Meters	RS-485 / Ethernet	Medium
LON (Lon Works)	Building automation	Twisted pair	Medium
KNX	Lighting / Blind / HVAC	TP, IP, RF	Medium
DALI	Lighting control	2-wire bus	Fast

- DDC (Direct Digital Control): Microprocessor-based controllers at field level
- Supervisory Level: BMS graphical interface, trending, alarm management
- Cloud Integration: API connections to smart metering, weather data, predictive maintenance

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Energy Efficiency

ASHRAE 90.1, LEED & Green HVAC Design

ASHRAE 90.1 Key Requirements

- Economiser Mode: Free cooling when OA enthalpy < RA enthalpy (climate zones 1–8)
- Supply Air Temperature Reset: Reset SAT up when zone demand is low
- Chiller Plant Efficiency: Min COP 5.5 for water-cooled centrifugal > 528 kW
- Boiler Efficiency: Min 80% Et for gas-fired > 88 kW input
- Fan Power Limitation: Max 1.25 W/L/s for supply fan (variable volume)
- Pipe & Duct Insulation: Per Table 6.8.3 — all supply/return must be insulated

Energy Benchmarks

Building Type	EUI (kWh/m²/yr)	HVAC Share	LEED Target (% vs baseline)
Office	150–220	40–50%	15–20% better
Hotel	250–400	45–55%	15–20% better
Hospital	400–800	50–60%	20–30% better
Retail	200–350	35–45%	10–15% better
Data Centre	600–2000	40–50% (IT+cooling)	PUE < 1.4

LOW-ENERGY HVAC STRATEGIES:

- Passive Cooling: Night purge ventilation, thermal mass, shading
- Radiant Cooling: Chilled beams / ceiling panels — lower supply temps, no fan coil noise
- TABS (Thermally Activated Building Systems): Concrete core cooling, lowest energy
- Geothermal Ground Loop: Stable source/sink temperature year-round
- Renewable Integration: Solar PV → electric HP; Solar thermal → DHW/absorption cooling

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Equipment Selection

Specifications, Performance & Procurement

AHU Selection Checklist

- Airflow (L/s): Sum of all zone peak flows with 10% spare
- External Static Pressure: Include all components + distribution system resistance
- Coil Entering/Leaving Conditions: Match psychrometric design state points
- Filter Grade: MERV-8 pre-filter + MERV-13 final (or HEPA for healthcare)
- Fan Class: Class I (<750 Pa), Class II (750–1500 Pa), Class III (>1500 Pa)
- Sound Power Level: Specify max dB(A) at each octave band
- Access for Maintenance: Min. 600mm clear access for coils, filters, fan

Chiller Plant Design

- N+1 Redundancy: Install n+1 chillers for 100% capacity with one standby
- Chilled Water ΔT: Design for 6–8°C (5.5°C supply, 11.5°C return) for efficiency
- Variable Primary Flow: Eliminates secondary pumps; differential pressure control
- Free Cooling Mode: Plate heat exchanger bypass when ambient ≤ 12°C (WC chiller)
- Thermal Storage: Ice or chilled water storage for peak demand shifting

PUMP SELECTION:

Head = System curve at design flow (all pressure drops summed)

Duty point: Pump curve intersects system curve at design flow

NPSH available > NPSH required + 0.5 m safety margin

Min. efficiency index (MEI) ≥ 0.4 (EU regulation)

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Commissioning & Maintenance

TAB, HVAC Cx & Preventive Maintenance

Testing, Adjusting & Balancing (TAB)

- Air Balancing: Measure and adjust supply/return airflows to design values (±10%)
- Hydronic Balancing: Proportional balancing method — start from index circuit
- Instrumentation: Pitot tube traversal, Capture hood, Ultrasonic flow meter
- Acceptance Criteria: ASHRAE 111 (duct), ASHRAE 90.3 (hydronic)
- Reports: Document all measured vs. design values for each terminal

Preventive Maintenance Schedule

Component	Monthly	Quarterly	Annually
Air Filters	Inspect / replace	—	—
Coils (evap/cond)	—	Clean & inspect	Full clean, UV treatment
Belts & Bearings	—	Tension check	Replace if needed
Refrigerant Charge	—	Leak check	Full performance test
Chiller Tubes	—	—	Water-tube brush clean
Cooling Tower	Biocide dosing	Blowdown check	Full inspection & clean
BMS Calibration	—	Sensor calibration	Full sequence check

COMMISSIONING STAGES:

1. Factory Acceptance Test (FAT) — verify before delivery
2. Installation Check — correct piping, ductwork, controls wiring
3. Pre-functional Tests — component-level checks (pump rotation, control signals)
4. Functional Performance Tests — full sequence of operations under load
5. O&M Training & Handover Documentation